1. A council of size 14 is to be selected from 10 women and 10 men. How many different councils can be selected if:

   a) There is no restriction?

$$\boxed{\binom{20}{14}}$$

d) subcouncil of size 5 is needed, with one president, one secretary and one treasurer

$$\binom{20}{14}\binom{14}{5}\cdot\binom{5}{3}\cdot 3!$$

   b) There must be equal number of women and men?

7w and 7M

$$\boxed{\binom{10}{7}\cdot\binom{10}{7}}$$

   c) There must be between 4 and 6 men?

M: 4 or 5 or 6

W: 10 or 9 or 8

$$\boxed{\binom{10}{4}\cdot\binom{10}{10}+\binom{10}{5}\cdot\binom{10}{9}+\binom{10}{6}\cdot\binom{10}{8}}$$

2. Consider alphabet 0, 1, 2, 3, 4, 5, 6, 7, 8 and strings of length 10.

a) How many strings have exactly three 4's?

$\frac{10!}{3!\,7!}$ 444 _ _ _ _ _ _ _ 8^7; 44 _ _ _ _ _ 4 _ _ 8^7; _ _ _ _ _ _ 4 _ _ 44 8^7

$$\boxed{\frac{10!}{3!\,7!}\cdot 8^7}$$

_ 4 _ _ _ 4 _ 4 _ _ 8^7

$$\boxed{\binom{10}{3}\cdot 8^7}$$

b) How many strings have exactly two 4's and five 3's?

$\frac{10!}{2!\,5!\,3!}$ 44333 33 _ _ _ 7^3; 34343 _ _ 3 _ 3 7^3

$$\boxed{\frac{10!}{2!\,5!\,3!}\cdot 7^3}$$

_ 4 3 3 3 4 _ _ 3 3

3 3 _ 3 4 _ 3 4 _ 3

7 positions for 3's & 4's
then select position for 4's ...

$$\boxed{\begin{array}{c}\binom{10}{2}\cdot\binom{8}{5}\cdot 7^3\\ \binom{10}{5}\cdot\binom{5}{2}\cdot 7^3\\ \binom{10}{7}\cdot\binom{7}{2}\cdot 7^3\end{array}}$$

c) How many strings have exactly six 1's and no other digits repeated?

1 1 _ 1 _ 1 1 1 _ _ (8, 7, 6, 5)

$$\boxed{\begin{array}{c}\binom{10}{6}\cdot 8\cdot 7\cdot 6\cdot 5\\ \binom{10}{6}\cdot P(8,4)\\ \binom{10}{6}\cdot\binom{8}{3}\cdot 3!\end{array}}$$

d) How many strings have at least seven 2's?

0	1	2	3	4	5	6	7	8	9	10
2	2	2	2	2	2	2	2	2	2	2

$$\boxed{\binom{10}{7}\cdot 8^3+\binom{10}{8}\cdot 8^2+\binom{10}{9}\cdot 8+1}$$

e) How many strings with exactly three 0's have no adjacent 0's?

8^7 ^1^1^1^8^5^1^7^ $\binom{8}{3}$; ^2^1^5^7^4^4^3^ $\binom{8}{3}$

$$\boxed{\binom{8}{3}\cdot 8^7}$$

↳ 7 other digits

_ 0 _ 0 _ 0 _

$x_1+x_2+x_3+x_4=7$

$\geq 0 \quad \geq 1 \quad \geq 1 \quad \geq 0$

$n=4;\ r=7-2=5$

$\binom{n+r-1}{r}$

$\binom{4+5-1}{5}=\binom{9}{5}$

$$\boxed{\binom{9}{5}\cdot 8^7}$$

3. What value does a variable *counter* have after the following code execution?

a

```
counter = 100

for i = 1 to 5 {
        counter = counter + 7
        for j = i+1 to 5 {
                counter = counter + 8
        }
}
```

$$counter = 100 + \underbrace{5}_{i}(7) + \underbrace{\binom{5}{2}}_{i,j}(8)$$

b

```
counter = 100

for i = 1 to 10 {
        counter = counter + 7
        for j = i+1 to 10 {
                counter = counter + 8
                for k = j+1 to 10 {
                        counter = counter + 9
                }
        }
}
```

for i = 1 to ~~10~~ 8 { … 9 … for j = i+1 to ~~10~~ { … 10 … for k = j+1 to ~~10~~ {

$$8 \qquad \binom{9}{2} \qquad \binom{10}{3}$$

$$counter = 100 + \underbrace{10}_{i}(7) + \underbrace{\binom{10}{2}}_{i,j}(8) + \underbrace{\binom{10}{3}}_{i,j,k}(9)$$

c

```
counter = 100

for i = 1 to 500 {
        counter = counter + 7
        for j = i+1 to 400 {
                counter = counter + 8
        }
}
```

* no gap

$$counter = 100 + \underbrace{500}_{i}(7) + \underbrace{\binom{400}{2}}_{i,j}(8)$$

d

```
counter = 100

for i = 1 to 400 {
        counter = counter + 7
        for j = i+1 to 500 {
                counter = counter + 8
        }
}
```

* gap

pairs missing! 401, 402, …, 500 missing pairs

$$counter = 100 + \underbrace{400}_{i}(7) + \underbrace{\left[\binom{500}{2} - \binom{100}{2}\right]}_{i,j}(8)$$

e

```
counter = 100
a = − 300
for i = 1 to n {
        counter = counter + 7
        a = i
}
for j = a to 2n {
        counter = counter + 8
        for k = j+1 to 2n {
                counter = counter + 9
        }
}
// assume n>1
```

a = n

for j = ~~a~~ n to 2n {

combinations

no gap

$j \sim 2n - n + 1 = n + 1$

pairs of $n, n+1, \ldots, 2n$

$$counter = 100 + \underbrace{n}_{i}(7) + \underbrace{(n+1)}_{j}(8) + \underbrace{\binom{n+1}{2}}_{j,k}(9)$$

$$x_1 + x_2 + \cdots + x_n = r \qquad \binom{n+r-1}{r}$$
$$\geq 0 \quad \geq 0 \quad \quad \geq 0$$

Review lecture topic "Combinations with repetition allowed". Then do the exercises:

4. In how many ways can a student distribute 10 identical marbles into six distinct containers such that:

   a) Every container has at least one marble?

$n = 6$ $\qquad \binom{n+r-1}{r}$

$r = 10 - 6 = 4$

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = \cancel{10}$$
$$\cancel{\geq 1} \quad \cancel{\geq 1} \quad \cancel{\geq 1} \quad \cancel{\geq 1} \quad \cancel{\geq 1} \quad \cancel{\geq 1}$$
$$\geq 0 \quad \geq 0 \quad \geq 0 \quad \geq 0 \quad \geq 0 \quad \geq 0$$

$$= \binom{6+4-1}{4} = \boxed{\binom{9}{4}}$$

   b) The last container has 5 or more odd number of marbles.

$\binom{n+r-1}{r}$

$x_6 = 5$ or $x_6 = 7$ or $x_6 = 9$

$r = 10 - 5 = 5$ $\qquad r = 10 - 7 = 3$ $\qquad r = 10 - 9 = 1$

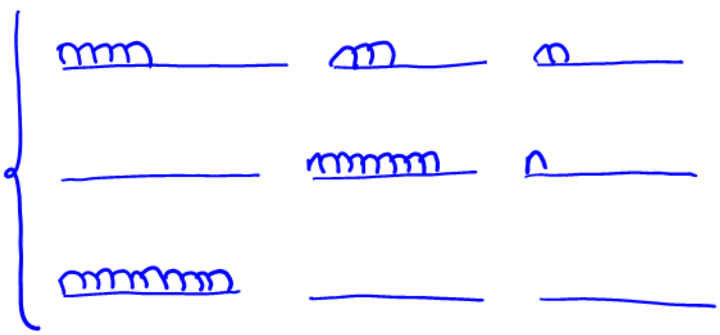
$n = 5$ $\qquad n = 5$ $\qquad n = 5$

$$\boxed{\binom{5+5-1}{5} + \binom{5+3-1}{3} + \binom{5+1-1}{1}}$$

5. In how many ways can a student arrange 10 distinct books on three shelves? Shelves could be empty. Books are placed next to each other and left to right with no empty spots between them.

$n = 3$; $r = 10$

$\binom{n+r-1}{r} = \binom{3+10-1}{10} = \binom{12}{10}$



$10!$

$10!$

$10!$

$$x_1 + x_2 + x_3 = 10$$
$$\geq 0 \quad \geq 0 \quad \geq 0$$

$$\boxed{\binom{12}{10} \cdot 10!}$$

If all of the problems can't be done in the lab period, students can work on them on their own.